Reliability - a simple concept helpful for understanding both simple and complex measurements

Timothy P. Hofer[1,2,*]

2025-09-20

Reliability is the most commonly reported measurement characteristic used as evidence of the quality of a measurement. However, researchers frequently miss opportunities to design studies in a way that allows reliability to be calculated. When reliability is reported, researchers often make a calculation that is not germane to the purpose of the measurement. Reliability can be described as a number that quantifies the ability of a measurement to distinguish between the members of a population (or the *objects of measurement*) with regard to a measured quantity. In this paper I review the technical definitions of reliability and describe the specifications of measurement purpose and conditions that are required in order to calculate the reliability relevant to the measurement task and report it in a way that is interpretable and more likely to be of use to other researchers. Reliability is discussed in the context of a well developed framework that provides a simple conceptual explanation for how reliability is defined and calculated but which allows for extension to more complex measurements and measurement study designs. This framework, known as generalizability theory, builds on classical test theory but has been developed and used widely over the past fifty years. With some exceptions, the research literature on quality and safety does not take advantage of it. Reliability is an important part of evaluating and improving the measurements that form the foundations of scientific research in health care quality and safety. This paper is designed to help researchers understand how to identify signal and noise correctly, specify the use of a measurement and consequently understand sources of measurement error better and thus be able to improve their measurements.

[1] Division of General Internal Medicine, Department of Internal Medicine, University of Michigan Medical School, Ann Arbor, MI, USA
[2] Center for Clinical Management Research, Veterans Affairs Ann Arbor Health System, Ann Arbor, MI, USA

[*] Correspondence: Timothy P. Hofer <thofer@umich.edu>

What is already known

- Challenging measurement problems are at the heart of efforts to improve quality and safety.
- Reliability is an important concept in understanding these measurement problems that has a clear technical definition and a strong theoretical and empirical basis.
- The concept of reliability has evolved with advances in measurement theory and analytic approaches over the past 50 years into a remarkably coherent framework for understanding and evaluating measurement called generalizability theory which is not widely known in the health care field.

What this study adds

- This paper first describes the concept of reliability and its formal definition and why it is useful.
- Using the modern measurement theory framework mentioned above, the paper identifies four key things to know about reliability to avoid the most common problems in the calculation of reliability and its interpretation.
- Additional detail and references necessary for developing more proficiency in the use of the framework supported by generalizability theory is provided in four appendices.

How this study might affect research, practice or policy

- Measurement is critical to evaluating the quality of health care and monitoring patient safety and is used extensively by managers and as part of national policy initiatives.
- The goal of these measurements is to distinguish between providers, or between time periods, with respect to the quality and safety of care that is delivered.
- Reliability provides a quantification of the ability to do that and a way to describe the degree to which these policies will be likely to succeed and are fair.

Introduction: The concept of reliability and why it is useful

> *Reliability* is a number that quantifies the ability of a measurement to distinguish between the objects of measurement (or members of a population) with regard to a measured quantity.

The ability of a measurement to distinguish between objects is determined by a balance between: 1) the *signal* in the measurement representing the distance between the true values in the population, and 2) the *noise* or measurement error. Noise in this formulation is composed of everything that is not signal. As a quantification of that balance, reliability can also be expressed as a function of the signal to noise ratio for a measurement.

In medicine, common measurements often require the use of an instrument such as a blood pressure cuff, a pulse-oximeter or clinical lab instrument. Measurement may also be done using an expert clinician rater who examines a patient, a medical record, pathology specimen, or imaging study. In other settings a set of survey items are combined into a scale used to measure patient reported functional status (such as the SF-36) or quality of life. But the same principles in quantifying reliability apply to all of these measurements.

We will use three exemplars of common measurements in research on healthcare quality and safety throughout this paper. Blood pressure is a clinical outcome measurement that is frequently used in clinical decision-making, such as diagnosis or whether to add additional medications, and as a component of quality metrics for providers and organizations. Adverse events (and sometimes preventable adverse events), as assessed by expert clinician raters who retrospectively review medical records, are commonly measured as part of patient safety studies. Finally, patient satisfaction, as measured by scales created by combining survey items, is one metric used in assessing the quality of care provided by clinicians, sites of care or health systems.

> Key point
>
> Characterizing what is signal and what is noise is the fundamental challenge of calculating reliability.

These three measurements can be used for different purposes, they can be measured with specified elements or randomly sampled elements from relevant populations of instruments, conditions or raters and applied to different populations of people. The calculation of reliability changes in important ways to reflect these differences. The basic principle remains that reliability is a simple function of signal and noise. The challenge is to identify those two components for the proposed use of a measurement. We do not cover how measurement error impacts analysis or statistical methods to mitigate measurement error as these are covered extensively elsewhere.[1,2] This paper is designed to help researchers understand how to specify the use of a measurement, identify signal and noise correctly and consequently understand sources of measurement error better and thus improve their measurements.

For the purposes of the following discussion, *measurement* refers to "placing objects into mutually exclusive and jointly exhaustive categories to which numbers are assigned based on a discernible attribute of the objects." (see Jacoby(1999),[3] also for earlier forms of this definition of measurement see Hand(1996)[4] and Campbell(1920)[5]) This definition includes continuous measurements as well as measurements that involve categorization into two or more groups, such as making a diagnosis or measuring the occurrence of complications, infection or errors. Given these definitions,

it is easy to appreciate how the concept of reliability is central to the purpose of measurement. How can you place objects into mutually exclusive and jointly exhaustive categories if you can't distinguish between them?

Four key things to know about reliability

There are problems in how reliability is presented in the literature. First, there are a lot of reliability statistics in common use with formulas that obscure the importance of identifying what is signal and what is noise in a measurement. This provokes unnecessary arguments and anxiety about which statistic to use and ultimately undermines the use of reliability to improve measurement. Second, researchers sometimes do not pick the correct calculation for the intended use of the measurement. Researchers also do not routinely specify in sufficient detail the conditions of measurement (populations of instruments, raters, measures, survey items, subjects) to which the measurement can be generalized and for which the calculated reliability is relevant. Finally as a summary measure, reliability may not be the best statistic to describe a measurement. Researchers should focus on learning four things about reliability to prevent these problems.

1. There is a simple approach to calculating reliability

The technical definition of reliability

There is a well-established technical definition of reliability that was mentioned in the literature back to the early 1900s [6] and formalized in the psychometric field in the 1960s.[7,8] This definition has a strong theoretical and empirical basis and a number of important practical implications that depend on that definition. A number of other statistics are loosely labeled as reliability or interpreted as reliability, some of which are not consistent with the established technical definition. Researchers should focus on reliability calculations that fit this definition.

> Key point
>
> Technical definition: Reliability is the proportion of variance in the observed measurement that is accounted for by the *true value*, where the true value is the mean over independent replications of the measurement

This definition arose in reference to the classical measurement theory model represented as follows:

$$X \text{ (measured value)} = T \text{ (true value)} + \epsilon \text{ (measurement error)}$$
$$\text{with reliability } R = \frac{\sigma_t^2}{(\sigma_t^2 + \sigma_\epsilon^2)}$$

This quantity may not obviously represent the concept of reliability as discussed above. However, σ_t^2 is the true variance or signal that we wish to detect. Similarly σ_e^2 is the noise in our measurement and $\sigma_t^2 + \sigma_e^2$ is the total observed variance. Reliability can be expressed as a function of the signal to noise ratio. Replicate measurements on a sample from a population of the objects of measurement are required to calculate reliability. A replicate measurement is one done under the same conditions of measurement such as two or more blood pressure measurements per person with a constant

expected true value. However, while two or more measurements per object of measurement as part of a measurement study are required to be able to calculate reliability, it is important to appreciate that the quantity above represents the reliability of using a **single** measurement made on each member of a similar population to distinguish between them (as opposed to averaging over 2 or more measurements for each person).

What is the "true value"

The true value represented by T can be a troublesome concept. What exactly do we mean by true? It may be possible to argue that a *gold standard measurement* exists and provides the "true" value of the entity we want to measure which is often referred to as the *target of measurement*. For example the International Bureau of Weights and Standards defines a metre by the length of a path traveled by light in a defined (very small amount of) time determined by an atomic clock and this could be considered close enough to the true value to be a gold-standard. One could imagine calculating the reliability of height measured using the stadiometer (the measuring stick) on a clinic scale where we also have research access to a true measured value made with an instrument that uses the Bureau of Standards method.

The comparison of a new or alternative measurement to an accepted gold-standard measurement, or at least a consensus reference standard (often informally referred to as a "bronze-standard") is best done with a different approach referred to as a "method comparison" study that is distinct from the framework used here.[9] For dichotomous measurements with a gold-standard, method comparison studies provide the familiar *misclassification statistics* such as sensitivity, specificity and postive and negative predictive values. Method comparison studies are briefly described in in Appendix 1 but are not discussed further here.

There is no gold standard for most measurements in the quality and safety literature. Commonly reported outcomes like the quality of various subsets of care (surgical, diabetes care, stroke care), "adverse events" preventable or not, "errors" of various types and even blood pressure have no widely accepted gold-standard. For the more common situation where there is no gold-standard, the true score represented by T should be thought of as a *true measured value* or, as specified in the definition of reliability above, the mean over a potentially infinite independent replications of the measurement. Of course this "true measured value" only has relevance when it is accompanied by a successful argument, supported by evidence, for the *validity* of the measurement.

The relationship of reliability and validity

One way to think of these two aspects of the quality of measurement is that generalizing from an observed measurement to the target of measurement (or thing we want to measure) requires two steps as shown in Figure 1. First we have to generalize from the observed measurement to the true measured value. Reliability calculations based on measurement studies provide evidence supporting this generalization. Then we have to make an argument that the true measured value represents the target of measurement or what we intended to measure. The evidence provided in this argument supports this second generalization step and thus the validity of the measurement.

An important point that has been made about reliability and validity is that efforts to improve reliability can reduce the validity of a measurement to the extent that changes made in the measurement to improve reliability weaken the connection between the true measured value and the

the target of measurement. (See further discussion of this in the conditions of measurement section and [7,10]. Another way to explain the distinction between reliability and validity is that reliability describes the ability to generalize over replicate measurements using the *same method* whereas validity describes how well we can generalize over measurements made using *different methods* intended to measure the same thing. While beyond the scope of this paper, a good discussion of the history of the concept of validity, debates about the existence of true values and how to develop a validity argument is provided by Strauss and Smith(2009)[11].

Variance components for calculating reliabilities can be estimated using multilevel models

Reliability is often calculated by choosing from a seemingly infinite number of special purpose statistics, usually named after someone (Cronbach, Armor, Krippendorf, Kuder-Richardson, Fleiss) or by a Greek letter (alpha, omega, phi, kappa) with formulas that may have no obvious relationship to the technical definition of reliability above. As a number of measurement experts have argued, many of these statistics are redundant or represent something other than reliability.[6,12–16] The plethora of names suggests an overly complex and amorphous concept. The simpler and more transparent approach using the technical definition of reliability can be carried out using a single measurement study with the appropirate number of repeated measurements when analyzed by newer analytic methods such as multilevel modeling, that make it easier to provide estimates for the variance components for making the calculation. Generalized linear latent and mixed models (GLLAMMs) estimated by maximum likelihood or Bayesian methods can be used for more complicated measurement study designs.[17–19] See appendix 2 for a more detailed description of the multilevel model for estimating the components of the reliability calculation.

Useful relationships that can be derived from the technical definition of reliability

Apart from the simplicity of using the technical definition to calculate reliability, a number of useful relationships involving this reliability can be derived when a calculation follows this definition as shown in Table 1.[20–23]

Table 1

Useful relationships involving reliability	Formula
1. reliability is the correlation between replicate measurements (Allen,2002)	reliability $R_x = r_{XX'}$
2. The signal to noise ratio (S/N) is a function of reliability (rearrangement)	S/N $= \frac{R}{1-R}$
3. The standard error of measurement (SEM) of X is a function of the reliability and the standard deviation of the observed X (Allen,2002)	SEM$_X = \sigma_x\sqrt{1-R_x}$
4. The reliability of an average of n replicate measurements is a function of the reliability of one (Spearman & Brown, 1910)	$R_n = \frac{n*R}{1+(n-1)*R}$

Useful relationships involving reliability	Formula
5. The true correlation is larger than the observed correlation when two variables X and Y are measured with error by an amount that is a function of their two reliabilities (Allen,2002)	$r_{T_X T_Y} = r_{X^* Y^*} \frac{1}{\sqrt{R_x R_y}}$
6. Rule of thumb: The effective sample size is less when using a variable measured with error by a factor equal to the reliability (Kraemer,2014)	$N_{effective} = N_{\text{using } x} R_x$
7. The reliability of a measurement is the same as the r-squared of the measured variable with its true score. This represents the maximum possible r-squared of that variable with any predictor, as nothing can predict X better than its true value (Allen,2002)	$R_x = r_{XX'} = r^2_{XT}$
8. The observed variance is inflated by measurement error and can be corrected by multiplying by the reliability, which is always less than or equal to 1, to obtain the true variance (substitution and rearrangment)	$\sigma_t^2 =$ observed $\sigma_x^2 \times R$

For example, based on a measurement study involving multiple blood pressure measurements of subjects in different settings by Llabre [24], the variance of the true systolic blood pressure when measured over a day at home and work is 144 mmHg and the variance of the noise is 238 mmHg. By the defintion above, the reliability then is 144/(144+238) or 0.38. Relationship 1 confirms that the correlation of two (or more) replicate measurements is the reliability of a single measurement. Relationships 2-3 give other ways to present the information contained in reliability and in 4 gives a formula to calculate a reliability for an operational measurement composed of an average of two or more replicate measurements of each entity as a way to improve measurement quality. Relationships 4-8 show ways in which low quality measurements undermine our ability to make clinical and policy decisions. These relationships provide a useful collection of information that can help inform the use of specific measurement. A specific worked out example using these relationships to inform a measurement is shown in Table 2. Other statistics that do not fit the technical definition or reliability do not allow us to describe these practical implications.

2. The reliability calculation changes with the purpose and conditions of measurement

The second thing to know is that there are many reliability calculations that can be made using the same measurement study. Which reliability calculation is appropriate depends on: 1) the purpose of the measurement, 2) whether measurements are combined by averaging in order to create the operational form of the measurement and 3) the conditions under which the measurement was made. It is meaningless to report a reliability without specifying those three things. (See also published guidelines on reporting measurement properties making a similar point [25,26]). This could seem to be an undesirable complication as compared to blithely using a familiar off-the-shelf reliability coefficient with an important sounding name without specifying these elements. But the use of any reliability coefficient has these elements embedded within its calculation and so to ignore them is to make it almost impossible to interpret the number and obscures important information from those who might use it. However the advantage of the framework presented here is that the

different reliability calculations can often be constructed from the same measurement study once the purpose, number of measurements and conditions of measurement are specified.

The measurement purpose determines the objects of measurement

Defining the purpose of measurement determines what should constitute the signal or variance of the true value in the numerator of the reliability calculation. Or as Bloch & Norman[27] put it, the first step is to answer "What is the object of measurement?." As an example, a survey-based measurement of patient satisfaction would commonly have three different purposes for a health care manager in a large health system.

1. How well can I distinguish between members of my patient population with respect to their satisfaction with my services? The purpose is to distinguish between patients and the patients are the objects of measurement.
2. Can I use this measurement for the purpose of distinguishing between primary care physicians with respect to the satisfaction of their patients. Here the purpose is to distinguish between primary care physicians, not patients.
3. Is average patient satisfaction a sufficiently reliable performance measure to distinguish between the medical directors who lead my 35 primary care locations? In this case the objects of measurement are the 35 primary care locations and the purpose is to distinguish between them.

Each of these purposes of a measurement has a *different* variance component that goes in the numerator of the reliability calculation and which represents the signal that is relevant to the ability to distinguish between the objects of measurement. The first requires the variance of the true measured satisfaction between patients in the population. The second requires the variance of the average satisfaction scores across primary care providers. The third requires the variance of the average satisfaction across the clinic locations in the health system. But a single measurement study across the 35 primary care locations with multiple patient respondents clustered within providers can provide the variance components necessary to carry out all three reliability calculations.

> Key point
>
> Using the same measurement for a different purpose can completely change the reliability calculation.

It is very common to see researchers offer a statistic that quantifies the reliability suitable for the first purpose of measurement listed above as a justification for the second and third purposes, when the magnitude of the former may have nothing to do with the magnitude of the appropriate reliabilities for the latter two purposes. For example, a patient satisfaction scale typically has a reliability for distinguishing between patients of 0.7-.95 (commonly reported using a staistic named Cronbach's alpha coefficient).[28] The 35 primary care locations that are the responsibility of the manager described above may have no appreciable differences in their average patient satisfaction in which case the variance and thus the calculated reliability with this variance in the numerator may be close to 0. The same patient satisfaction survey can easily be reliable for one purpose and not for another. A second example involving adverse events is given in Table 3.

Averaging two or more measurements improves reliability

The reliability calculation also depends on the number of measurements that are averaged to produce an operational measurement. As illustrated in Table 2, we were able to use relationship 6 from Table 1 to calculate that the average of 6 home BP measurements has a reliability of 0.80 where one measurement only had a reliability of 0.40 for distinguishing between people with respect to their blood pressure. This same calculation can be made more transparently using the technical definition of reliability given above. The formula can be extended to allow calculation of the reliability of an operational measurement when it is composed of an average of replicate measurements simply by dividing the noise or error term by n_i, which is the number of BP measurements per person.

$$\text{reliability} = R_{n_i} = \frac{\sigma_t^2}{\sigma_t^2 + \frac{1}{n_i}\sigma_\epsilon^2}$$

> Key point
>
> Measuring more than once is the easiest way to improve the reliability of a measurement.

An implication of this is that reporting a reliability requires not just a statement defining the purpose but also the number of measurements that are averaged to produce the operational measurement that will be used. For example:

> We calculated a reliability of 0.80 for the purpose of distinguishing between patients with respect to blood pressure when measured using an average of 6 measurements.

A more detailed example involving a patient satisfaction survey is shown in Table 4. For a survey scale we consider the separate survey items to be multiple measurements that are combined to form the patient satisfaction scale. This is an example where the operational measurement is always the sum or average of the component items of the scale and not a single measurement.

Defining the conditions of measurement requires identifying the sources of variation.

Up to this point we have focused on a measurement model where the denominator of reliability has one undifferentiated error term σ_ϵ^2. One of the more important developments in measurement theory since the classical model has been based on introducing an expansion of the undifferentiated error term to reflect a combination of the major sources of variation. This extension to classical formulations of measurement error is known as *generalizability theory*. (See Appendix 4 and[7,10,29]). When each operational measurement event occurs using a potentially different randomly sampled instrument, rater, scale item or time period from the respective populations of each, then the variation in these conditions contributes to the variation in the observed value of the measurement. Specifying the conditions of measurement requires identifying these sources, populations and sampling procedures. This is what determines the components of the reliability calculation that form the noise term in the denominator of the reliability calculation shown below.

$$\text{reliability } R = \frac{\sigma^2_{\text{signal}}}{\sigma^2_{\text{signal}} + \underbrace{\sigma^2_{\text{source-1}} + \sigma^2_{\text{source-2}} + \sigma^2_{\text{residual}}}_{\text{Noise}}}$$

Reliability can be improved by 1) improving the measurement by making some of the components of noise smaller, say by additional training or instrument calibration or 2) standardizing the measurement condition by fixing a source of variation, such as using only a limited number of well calibrated research instruments for measuring blood pressure (as discussed in Table 4 and in more detail in Table 5). With a *fixed condition* the variance component reflecting that source of variation, say $\sigma^2_{\text{source-2}}$ can be removed from the denominator. Both strategies come at a cost in generalizability of the measurement. If a rater or observer is highly trained then the measurement can be generalized only to situations where the rater or observer is similarly trained. If a measurement is made with a fixed set of research instruments, then it can be generalized to measurements made with those instruments and not others that might be used in usual clinical practice.

> Key point
>
> Characterizing and quantifying the major sources of noise or unwanted variation in a measurement is the critical first step to describing and improving the quality of that measurement.

Design of measurement studies to estimate key sources of variation and calculate reliability

The best design for a measurement study to estimate the magnitude of the different variance components is one where the objects of measurement are crossed with the conditions of measurement in a fully *factorial design*. So if a clinic blood pressure measurement was felt to vary on the basis of the machine and the operator as well as the patient, then you would want a study with 2 or more measurements for each combination of patient, machine and operator. With such a design, a single measurement study can be used to estimate any of the many reliabilities often given different names (and formulas) such as inter or intra-reliability, internal consistency, or test-retest reliability simply by fixing different conditions and number of measurements. See Appendix 4 for details and [10].

In health care it is much more common to find measurement study designs that are nested of necessity (it is not possible to systematically assign providers or patients to every hospital) or by pragmatic choice (it is difficult to repeat intrusive or invasive measurements). The rules about the inclusion and exclusion of elements in the error or noise term of the reliability calculation become more complicated with additional sources of variation and highly *nested designs*. If such complexity is required then the reader is advised to consult a number of primers on the topic. (See Appendix 4 and [10,27,29–31]). The estimation of the variance components themselves for more complex models is increasingly tractable using GLAMMs and Bayesian estimation methods as noted above.[17–19,32]

However, it is important to emphasize that just going through the exercise at a conceptual level by trying to identify the major sources of variation and how they contribute to the reliability calculation can give valuable insights about the noise or error term in a measurement and how it might be managed. This is true even if ultimately a simpler reliability calculation is done with an undifferentiated error term. But if not estimated separately or at least described, the major sources of variation will always still be there hidden in the residual error term. Whether the levels of the important sources of variation will be fixed or sampled and the populations from which any sampling was done must be described explicitly in the measurement procedure in order to support

generalizing the reliability to any other setting and to define a replicate measurement (See Table 5 for an example).

3. Reliability is a summary measure

It is also important to know that reliability is a summary measure combining several different elements relevant to distinguishing between the objects of measurement. As such it has all the advantages and disadvantages of a summary measure. Some questions about measurement are answered better by the elements that make up the reliability calculation described above, rather than by the summary measure. In particular, the square root of the measurement error variance is the *standard error of measurement* (SEM) as shown in Table 1 and demonstrated in Table 2. This quantity is often of interest and is in the same units as the measurement itself, unlike reliability which is a dimensionless number.

> Key point
>
> For some purposes, the components of reliability are more relevant than the overall summary measure.

The relative magnitude of the variance components that arise from the conditions of measurement may also be of significant interest. Because of the way measurement errors combine, if the standard error is 10 for one component and 1 for a second then the combination of the two produces a standard error of measurement of only $\sqrt{10^2 + 1^2} = 10.05$. A second source of variation that has a standard error that is 10% of the first only increases the combined standard error of variation by 0.5%. Knowing the relative size of the variance components is critical to knowing where to invest efforts to reduce measurement error.[33]

4. Reliability is specific to a population

The fourth, and possibly most important thing to know about reliability is that as with any statistic used to estimate a population parameter, the calculation is only generalizable to the population from which the samples were drawn to calculate the statistic. As an example, suppose that we have developed a patient satisfaction measurement that has a reliability of 0.8 for a task of distinguishing between patients with respect to their satisfaction based on a measurement study done in our health system. As part of the reliability calculation we estimated the standard deviation of the true patient deviations on a 60 point scale is 20 (σ_p) with a mean satisfaction of 30. We can also calculate that the standard error of measurement is 10 points (σ_e , calculated as shown in Table 1, relationship 3).

We might be pleased by this and decide to sell our satisfaction assessment system to another health system whose patients have higher and less variable true measured scores, let's say an average of 45 with true measured patient scores having a standard deviation of only 10. Without changing anything about our measurement procedure the reliability of our measurement system is only 0.50 in the new health system.

$$R_{system_1} = \frac{20^2}{20^2 + 10^2} = 0.80 \text{ but } R_{system_2} = \frac{10^2}{10^2 + 10^2} = 0.50$$

If you plan to use a measurement based on its high reported reliability, you must make sure that the population where you intend to use it is similar to the one in which the reliability was calculated. In particular it must have a similar variance of the true measured value for the measurement to have a similar ability to distinguish between the objects of measurement.

> Key point
>
> It is meaningless to say you are using a "reliable" measurement if you have not made an argument that your population is similar to that in which the reliability was calculated.

Sometimes researchers select samples for measurement studies in such a way so as to increase the variability of the target of measurement in the sample. For example, one might be tempted to oversample adverse events, which are otherwise rare, for the purpose of doing a study to calculate reliability. This increases the heterogeneity of the sample relative to the population of interest. Without accounting for the oversampling in the analysis, this will produce a reliability that is larger and not relevant to the population where the measurement will be used. A measurement study is most useful when done in a broad, representative and well characterized population so that others can argue for the generalizability of the calculated reliability to their population.

Dichotomous and ordinal measurements

The models above are designed for continuous data representing interval or ratio level measurements. What about binary and ordinal data? The most obvious way to quantify reliability is to use the same multilevel model approach but with a generalized linear model as shown in Appendix 2. This approach can provide estimates of the variance components needed to calculate a reliability. However a key difference from the calculations done for continuous data is that this reliability is calculated on the *latent scale* of the linear predictor (the log-odds scale in the case of a logistic regression) which is then transformed by the link function to the probability scale of the observed binary event.

Kappa statistic

There is an enduring and popular statistic for dichotomous and ordinal measurements called the kappa statistic. As noted by Kraemer [14], the original form of kappa was derived in 1960 from an ad hoc starting point as a chance corrected measure of agreement, and as such was simply a sample descriptive statistic. However, Kraemer was able to show that if one invokes the same measurement model as above, including the existence of a "true-measured value" that is the mean of a large number of replicate measurements with randomly selected raters, then the formula for kappa corresponds to one that exactly fits the technical definition of reliability.

Under these assumptions, kappa is interpretable as a population parameter and is often referred to as the intraclass kappa when used this way. Kraemer goes on to say of other statistics purporting to measure reliability of dichotomous measurements that "Unless such a statistic estimates the same population parameter as does the intraclass kappa, it is not an estimate of the reliability of a binary measure."[14] Kraemer also showed that kappa can be used in many of the same relationships shown in Table 1 for reliability derived from classical measurement theory.[21] While Cronbach himself described statistics like Cronbach's alpha as unnecessary with the advent of generalizability theory

(see Table 4), Kappa is still useful as it represents a reliability on the observed or manifest probability scale, which is otherwise challenging to calculate.[34]

Latent vs. Manifest scale

Both the latent and observed, or manifest, scale are of interest. The latent scale may be more relevant when talking about the underlying processes that are modeled and which we want to understand. We often conceptualize the integration of these processes on the latent scale as giving rise to the realized binary and ordinal outcomes. The manifest scale is more relevant when thinking about quantifying better or worse measurement in terms of the real world consequences like misclassification. Of course they would not both be of interest unless they were different, and they are. The level of reliability is always lower on the manifest scale for dichotomous measurements.(See Appendix 3 for further discussion of this distinction and its reasons.)

Some practical suggestions

There are unfortunately more choices to be made for estimating variance components and reliability calculations for binary and ordinal data as a result of the above discussion.

- If using a simpler model with an undifferentiated error term the intraclass kappa will produce a reliability that fits the technical definition of reliability on the observed or manifest scale.[6,14]
- With more complex models for binary data where the there are two (or more) major sources of error:
  - Reporting reliability of these type of data should where possible include estimates on both the latent and manifest scale with explicit arguments about which is the most relevant to the purpose of a specific measurement.
  - The more complex model and data can be simplified in different ways depending on the purpose of measurement allowing the calculation of an intraclass kappa to describe the appropriate reliability on the manifest scale.[14]
  - Variance components and reliability can and should be calculated as for continuous data using generalized linear models but the reliability is on the latent log odds scale and should be reported and interpreted as such.
  - If you have balanced designs you can use anova methods with bootstrapped standard errors to estimate variance components that allow you to calculate reliability on the manifest scale.[35]
- Ordinal data is probably best, or at least most easily treated as continuous and reliability calculated using ICCs as with continuous data.[35]

Conclusion

Reliability is a number (statistic) with which one can make inferences about a useful population parameter. It has a technical definition that is simple in concept and rich in practical uses but reliability is still often misunderstood. The concept has been obfuscated by the development of too many other statistics that are labeled or used as reliability statistics and which are at best redundant and at worse are able only to provide sample descriptions. As Geoffrey Norman said "Obscure

coefficients are like obscure composers; there's likely a good reason why they have faded into obscurity".[12] There are many reliability coefficients that can be estimated using the same instrument or measurement method. Estimating the correct parameter requires a precise specification of the purpose of measurement and the conditions of measurement including any averaging of replicate measurements. The reward is that the reliability can tell you a lot about the consequences of using the measurement procedure as designed. The valuable byproduct is that it forces investigators to be precise about the purpose of measurement as well as the population to which it is possible to generalize the measurement. Reliability is a key diagnostic tool for identifying the ways to improve a measurement.

References

1 Shaw PA, Gustafson P, Carroll RJ, *et al.* STRATOS guidance document on measurement error and misclassification of variables in observational epidemiology: Part 2—More complex methods of adjustment and advanced topics. *Statistics in Medicine*. 2020;39:2232–63. doi: 10.1002/sim.8531

2 Keogh RH, Shaw PA, Gustafson P, *et al.* STRATOS guidance document on measurement error and misclassification of variables in observational epidemiology: Part 1—Basic theory and simple methods of adjustment. *Statistics in Medicine*. 2020;39:2197–231. doi: 10.1002/sim.8532

3 Jacoby WG. Levels of Measurement and Political Research: An Optimistic View. *American Journal of Political Science*. 1999;43:271–301. doi: 10.2307/2991794

4 Hand DJ. Statistics and the Theory of Measurement. *Journal of the Royal Statistical Society Series A (Statistics in Society)*. 1996;159:445–92. doi: 10.2307/2983326

5 Campbell NR. *Physics - the elements*. Cambridge: Univ. Press 1920.

6 Streiner DL, Norman GR, Cairney J. *Health measurement scales: A practical guide to their development and use*. Fifth edition. Oxford: Oxford University Press 2015.

7 Cronbach LJ, editor. *The Dependability of behavioral measurements: Theory of generalizability for scores and profiles*. New York: Wiley 1972.

8 Lord FM, Novick MR, Lord FM, *et al.* *Statistical theories of mental test scores*. Reading, Mass.: Addison-Wesley Pub. Co. 1968.

9 Barnhart HX. A Review on Assessing Agreement. *Wiley StatsRef: Statistics Reference Online*. John Wiley & Sons, Ltd 2018:1–30.

10 Brennan RL. *Generalizability theory*. New York: Springer 2001.

11 Strauss ME, Smith GT. Construct validity: Advances in theory and methodology. *Annual Review of Clinical Psychology*. 2009;5:1–25. doi: 10.1146/annurev.clinpsy.032408.153639

12 Norman GR. Discrimination and reliability: Equal partners? *Health and Quality of Life Outcomes*. 2008;6:81. doi: 10.1186/1477-7525-6-81

13 Kraemer HC. Measurement of reliability for categorical data in medical research. *Statistical Methods in Medical Research*. 1992;1:183–99.

14 Kraemer HC, Periyakoil VS, Noda A. Kappa coefficients in medical research. *Statistics in Medicine*. 2002;21:2109–29. doi: 10.1002/sim.1180

15 Vach W. The dependence of Cohen's kappa on the prevalence does not matter. *Journal of Clinical Epidemiology*. 2005;58:655–61. doi: 10.1016/j.jclinepi.2004.02.021

16 Vach W, Gerke O. Gwet's AC1 is not a substitute for Cohen's kappa - A comparison of basic properties. *MethodsX*. 2023;10:102212. doi: 10.1016/j.mex.2023.102212

17 Vangeneugden T, Laenen A, Geys H, *et al.* Applying Concepts of Generalizability Theory on Clinical Trial Data to Investigate Sources of Variation and Their Impact on Reliability. *Biometrics*. 2005;61:295–304. doi: 10.1111/j.0006-341X.2005.031040.x

18 Laenen A, Vangeneugden T, Geys H, *et al.* Generalized reliability estimation using repeated measurements. *The British Journal of Mathematical and Statistical Psychology*. 2006;59:113–31. doi: 10.1348/000711005X66068

19 Jiang Z, Skorupski W. A Bayesian approach to estimating variance components within a multivariate generalizability theory framework. *Behavior Research Methods*. 2018;50:2193–214. doi: 10.3758/s13428-017-0986-3

20 Allen MJ, Yen WM. *Introduction to measurement theory*. Reissued. Long Grove, Ill: Waveland Press 2002.

21 Kraemer HC. The Reliability of Clinical Diagnoses: State of the Art. *Annual Review of Clinical Psychology*. 2014;10:111–30. doi: 10.1146/annurev-clinpsy-032813-153739

22 Spearman C. Correlation calculated from faulty data. *British Journal of Psychology, 1904-1920*. 1910;3:271–95.

23 Brown W. Some experimental results in the correlation of mental abilities. *British Journal of Psychology*. 1910;3:296–322.

24 Llabre MM, Ironson GH, Spitzer SB, *et al.* How many blood pressure measurements are enough? An application of generalizability theory to the study of blood pressure reliability. *Psychophysiology*. 1988;25:97–106.

25 Gagnier JJ, Lai J, Mokkink LB, *et al.* COSMIN reporting guideline for studies on measurement properties of patient-reported outcome measures. *Quality of Life Research: An International Journal of Quality of Life Aspects of Treatment, Care and Rehabilitation*. 2021;30:2197–218. doi: 10.1007/s11136-021-02822-4

26 Kottner J, Audige L, Brorson S, *et al.* Guidelines for Reporting Reliability and Agreement Studies (GRRAS) were proposed. *International Journal of Nursing Studies.* 2011;48:661–71. doi: 10.1016/j.ijnurstu.2011.01.016

27 Bloch R, Norman G. Generalizability theory for the perplexed: A practical introduction and guide: AMEE Guide No. 68. *Medical Teacher.* 2012;34:960–92. doi: 10.3109/0142159X.2012.703791

28 Satpathy S, Wundaville LT, Satapathy S, *et al.* A Systematic Review of Patient Satisfaction Scales and Their Applicability to Covid-19 Hospitalized Patients: Gaps and Emerging Needs. *Journal of Patient Experience.* 2022;9:1–10. doi: 10.1177/23743735221079132

29 Shavelson RJ, Webb NM, Shavelson RJ, *et al.* *Generalizability theory : A primer.* Newbury Park, Calif.: Sage Publications 1991.

30 Webb NM, Shavelson RJ, Haertel EH. 4 Reliability Coefficients and Generalizability Theory. In: Sinharay CRR and S, ed. *Handbook of Statistics.* Elsevier 2006:81–124.

31 Mokkink LB, Eekhout I, Boers M, *et al.* Studies on Reliability and Measurement Error of Measurements in Medicine – From Design to Statistics Explained for Medical Researchers. *Patient Related Outcome Measures.* 2023;14:193–212. doi: 10.2147/PROM.S398886

32 Rabe-Hesketh S, Skrondal A. Classical latent variable models for medical research. *Statistical Methods in Medical Research.* 2008;17:5–32. doi: 10.1177/0962280207081236

33 Kane M. The Errors of Our Ways. *Journal of Educational Measurement.* 2011;48:12–30.

34 Milanzi E, Molenberghs G, Alonso A, *et al.* Reliability measures in item response theory: Manifest versus latent correlation functions. *The British Journal of Mathematical and Statistical Psychology.* 2015;68:43–64. doi: 10.1111/bmsp.12033

35 Kraemer HC. Kappa Coefficient. In: Balakrishnan N, Colton T, Everitt B, *et al.*, eds. *Wiley StatsRef: Statistics Reference Online.* Wiley 2015:1–4.

36 Abel GA, Agniel D, Elliott MN. The good, the bad and the ugly: What do we really do when we identify the best and the worst organisations? *BMJ Quality & Safety.* 2025;34:53–61. doi: 10.1136/bmjqs-2023-017039

37 Hiatt HH, Barnes BA, Brennan TA, *et al.* A study of medical injury and medical malpractice. *N Engl J Med.* 1989;321:480–4.

38 Bravo G, Potvin L. Estimating the reliability of continuous measures with Cronbach's alpha or the intraclass correlation coefficient: Toward the integration of two traditions. *Journal of Clinical Epidemiology.* 1991;44:381–90.

39 Altman DG, Bland JM. Measurement in Medicine: The Analysis of Method Comparison Studies. *Journal of the Royal Statistical Society Series D (The Statistician)*. 1983;32:307–17. doi: 10.2307/2987937

40 Choudhary PK, Nagaraja HN. *Measuring agreement: Models, methods, and applications*. Hoboken, NJ: John Wiley & Sons 2017.

41 Wang W, Lin N, Oberhaus JD, *et al.* Assessing method agreement for paired repeated binary measurements administered by multiple raters. *Statistics in Medicine*. 2020;39:279–93. doi: 10.1002/sim.8398

42 Skrondal A, Rabe-Hesketh S. *Generalized Latent Variable Modeling: Multilevel, Longitudinal, and Structural Equation Models*. Boca Raton: Chapman & Hall/CRC 2004.

43 Mood C. Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It. *European Sociological Review*. 2009;26:67–82. doi: 10.1093/esr/jcp006

44 Buis ML. Logistic regression: Why we often can do what we think we can do. 2015.

45 Kraemer HC. Ramifications of a population model for κ as a coefficient of reliability. *Psychometrika*. 1979;44:461–72. doi: 10.1007/BF02296208

46 Rodriguez G, Elo I. Intra–class correlation in random–effects models for binary data - The Stata Journal Volume 3 Number 1, St0031. *Stata Journal*. 2003;3:34–46.

47 Psychological testing. *Wikipedia*. 2025.

48 Standard (metrology). *Wikipedia*. 2025.

49 Kane MT. A Sampling Model for Validity. *Applied Psychological Measurement*. 1982;6:125–60. doi: 10.1177/014662168200600201

50 Schielzeth H, Nakagawa S. Nested by design: Model fitting and interpretation in a mixed model era. *Methods in Ecology and Evolution*. 2013;4:14–24. doi: 10.1111/j.2041-210x.2012.00251.x

51 Brennan RL. Coefficients and Indices in Generalizability Theory. University of Iowa, Iowa City, IA: Center for Advanced Studies in Measurement and Assessment 2003, August.

Glossary

Agreement: Agreement does not have a single widely accepted technical defintion and is used to describe different things by different people. Those who use the generalizability framework discussed in this paper seem to define agreement as the standard error of measurement. Some of them then find it less essential to report as it can always be calculated from the elements of the reliability calculation. Some argue it is an important quantity to routinely provide pointing out that often reliability is given without specifying its components, making the standard error of measurement difficult to caclulate.

Others reserve the term agreement for the specific case of method comparison studies (discussed in Appendix 1) in which case it is used to label a measure of association between a new measurement and gold standard, or between two measurements of the same thing made with different methods. The kappa statistic can be used in this way as described in Appendix 3.

Beyond these two specific uses, the word agreement gets used in other contexts including as a synonym for calculations that fit the technical defintion of reliability as well as with simply its common language meaning of consistency or lack of incompatibility.

Generalizability theory: A framework for calculation reliability that is distinguished by breaking apart the undifferentiated measurement error term that appears in the classical test theory model into component parts determined by the major sources of variation in a measurement. This decomposition of measurement error allows researchers to identify the largest contributors to the measurement error which then can suggest ways of improving the measurement.

Factorial design: A measurement study design in which the objects of measurement are fully crossed with the conditions of measurement with two or more measurements within each cell.

Fixed condition of measurement: An approach to improving measurement that involves limiting the instruments, raters or survey items used in a measurement to a fixed specific set from among all the possible elements that might be used to make a measurement. This improves the reliability of the measurement by removing the sources of variation associated with allowing the measurement to be made by randomly sampling from populations of instruments, raters or survey items. However, this also correspondingly limits the relevance of any reliability calculation to those fixed elements. The reliability of measurements done using other instruments raters or survey items is then unknown.

Gold standard measurement: A measurement that produces the true value.

Measurement: A process of placing objects into mutually exclusive and jointly exhaustive categories to which numbers are assigned based on a discernible attribute of the objects.

Noise: Everything that is not signal, the unwanted variation in a measurement that functions to hide or obscure the signal.

Latent scale: For dichotomous and ordinal measurements analyzed using generalized linear models the analysis is carried out on the latent scale of the linear predictor. It is easiest to calculate reliability on this scale. The text discusses the issues in interpretation of the latent vs the manifest scale.

Manifest scale: This refers to the observed dichotomous and ordinal scale of a measurement when the measurement has those levels of measurement.

Measurement procedure: The process by which a number is applied to the objects of measurement. The appropriate use of a measurement depends on a clearly communicated or understood *measurement procedure.*[4] **Misclassification statistics:** These include most commonly, sensitivity, specificity, as well as positive and negative predictive values.

Nested design: A measurement study design in which some elements of a condition of measure-

ment occur only within specific elements of a second condition of measurement. A common example would be that there are not many cases where a patient would have have an operation with each of 2 or more surgeons. In that case, the design has patients nested within surgeon. Similarly patients would generally report satisfaction only for one provider or clinic and would not be able to provide measurements for two or more of either.

Objects of measurement: The collection of objects to which a number is assigned.

Reliability: a number that quantifies the ability of a measurement to distinguish between the objects of measurement.

Signal: That part of the variation in a measurement that is of interest given the purpose of measurement.

Standard error of measurement: This is the square root of the measurement error variance represented by the noise defined here as all the terms that are not signal in the reliability calculation. This gives a sense on average of how the repeated measurements of a single object of measurement will vary.

Target of measurement: The entity which the measurement is intended to quantify.

True measured value: The expected value over potentially infinite independent replications of the measurement.

True value: The value of a measurement of the intended concept of interest in the absence of any systematic bias or random noise.

Validity: A successful argument that the measurement represents the intended concept of interest or target of measurement.

Figure 1 caption

Reliability and validity as two distinct steps in generalizing from the observed measurement to the target of measurement.

Box explanations

Table 2: Using reliability identities to inform blood pressure measurement

Assume that the reliability is approximately 0.40 for the purpose of distinguishing between the subjects in our clinic population with respect to their blood pressure when using a single randomly selected systolic blood pressure (SBP) measurement done with an ambulatory blood pressure monitor at home and work over single day (as found in Llabre(1988) [24]).
Using the relationships from Table 1 we can calculate:

1. that the correlation between repeated SBP measurements at home is 0.40.
2. the signal to noise ratio is 2/3 or two parts signal to three parts noise.
3. If our clinic population has an observed standard deviation of 20mm Hg in their home blood pressures, the standard error of measurement for a home SBP value is $20 * \sqrt{1 - .4} = 15$ mm Hg. Thus if a patient has a measured value of 140 mm Hg the range of 140 +/- 30 would include their true measurement 95% of the time. This is not very useful for clinical decision making about treatment as the confidence intervals include quite low to quite high SBPs for which the treatment decisions would be different.
4. With 6 home SBP measurements the reliability of the average is $R_6 = \frac{6*0.40}{1+(6-1)*0.40)} = 0.80$
5. If we found as part of another investigation that the correlation of SBP and cholesterol (measured by the laboratory with reliability 0.98) done at the last clinic visit was 0.3, then we can caclulate that the correlation of the true values measured without error would be about 0.5 ($\frac{0.3}{\sqrt{.4*.98}} = 0.49$), substantially higher than originally estimated.
6. A study looking for a moderate effect size difference in SBP between groups that would require 84 subjects if we had no measurement error would require approximately 2.5 times the number of subjects using our home SBP measurements (84/0.4).
7. If we wanted to build a prediction model of home SBP, we could explain at most 40% of the variation even with a perfect prediction model.
8. If the mean SBP of our clinic population is 140 the health system managers could call us to task for having a high proportion (16%) of our population with a high risk SBP over 160 (based on the inflated observed standard deviation of 20mmHg for our population). In fact the proportion of patients with the true SBP over 160 is only 5-6% based on the true standard deviation of $\sqrt{20 \text{ mm Hg}^2 \times (0.40)} = 12.5$mmHg. If estimating medical costs, the inflation of observed variance will lead you to overestimate of the range of costs attributable to patients and the savings that could be achieved by lowering the cost of high outliers to average. For a dichotomous variable this inflation will lead to misclassification, for example, of the diagnostic categories into which patients are placed or an overestimate of a rate of adverse events or poor performing hospitals.[13,36]

Table 3: A change in the purpose of measurement changes the reliabilty calculation

The reporting of adverse events is another instructive example for how the purpose of measurement should change the calculation of reliability. The famous Harvard Medical Practice Study (HMPS), which helped to launch the patient safety movement, was actually motivated by an interest in evaluating the feasibility of tort reform that proposed compensating patients for all adverse events rather than litigating whether they were negligent or not.[37] In order to compensate patients fairly you would need to be able to distinguish between them as to whether they suffered an adverse event or not. Thus the ability of the measurement to distinguish between patients would be an appropriate reliability to report so as to help understand the feasibility of the tort reform proposal. The numerator is the variance between patients.
When the results of the HMPS were re-purposed to support a new patient safety movement, adverse events (AEs) became a primary outcome for measuring and monitoring patient safety. In this context, the purpose of measurement is different. There is little evident interest in identifying particular patients who suffered an AE. Rather the purpose is more commonly to distinguish between hospitals or health systems with respect to the safety of care as measured over the patients they care for. Or the purpose may be to distinguish between time periods seeking evidence for whether adverse event rates are different over time either within or over hospitals. The numerator for a reliability calculation for these purposes would require estimates of the variance between hospitals or health systems, or between time periods. These latter reliability calculations are not commonly reported. Most commonly the reliability for distinguishing between patients is the one reported. An example of a calculation of reliability for the purpose of using a measurement aggregated at a organizational level is presented later in Table 5.

Table 4: Measuring more than once

Patient satisfaction is commonly measured using a scale that is composed of multiple independent measurements of the same concept. As an example, suppose a health care manager decides to survey their patient population using a satisfaction scale composed of an average of 5 item responses administered to each person.

To calculate the reliability, the manager first specifies the purpose of measurement which is to distinguish between people with respect to their patient satisfaction and thus the numerator of the reliability calculation will be the person level variance σ_p^2 as estimated from a multilevel model with person as a random effect.

The manager might use a measurement strategy, similar to standardized testing in education, where for any given survey administration 5 items are selected from a large national item bank of different questions measuring patient satisfaction. As the reliability is intended to generalize to other surveys done with different samples of items from the national pool, the variation across patients in response to different items would contribute to the error or noise component of the reliability calculation which for a sum or average of 5 randomly sampled item measurements would have the following form:

$$\text{for a five item scale } R_5 = \frac{\sigma^2_{\text{person}}}{\sigma^2_{\text{person}} + \underbrace{\frac{\sigma^2_{\text{item}} + \sigma^2_{\text{person x item, }\epsilon}}{5}}_{\text{Noise}}}$$

However, more commonly the items in a survey scale are considered fixed and then the survey would always be administered using the same 5 items. By fixing the scale items the measurement is standardized in a way similar to a measurement strategy of using the same blood pressure instrument for every patient in a clinical study. When a source of variation is fixed it can be removed from the noise component of reliability and the calculation becomes.

$$\text{for a five item scale } R_5 = \frac{\sigma^2_{\text{person}}}{\sigma^2_{\text{person}} + \underbrace{\frac{\sigma^2_{\text{person x item, }\epsilon}}{5}}_{\text{Noise}}}$$

For this second calculation the item identifiers are included as dummy variables in the multilevel regression removing that source of variation. As noted above, the random effect for the person identifier gives the variance component for the numerator and then the residual is σ_e^2 in the denominator of the reliability calculation. The reliability calculated using this mixed effect model with patients as random and items as fixed for 5 measurements is a close analog of Cronbach's alpha.[38] Given the flexibility and transparency of this generalized approach to the calculation of reliability, Cronbach said of his eponymously named statistic that:

> I no longer regard the alpha formula as the most appropriate way to examine most data. Over the years, my associates and I developed the complex generalizability (G) theory which can be simplified to deal specifically with a simple two way matrix and produce coefficient alpha.

Table 5: The conditions of measurement for determining the occurrence of adverse events

Researchers measure adverse events during a hospitalization using a method that relies on the judgement of expert clinical raters who review medical records. For distinguishing between hospitals the numerator of the reliability calculation is the variance in hospital adverse event rates. Each rater reviews a random sample from all hospitals (not every record) and patients are nested within hospital. This is then a partially crossed and nested design. The noise term is expanded by identifying potentially important sources of variation.

- First, the patients effectively represent independent measurements of the hospital for this purpose and there is likely to be a large amount of variation across patients.
- Second, the raters may differ from each other with variance σ^2_{rater} in the overall average rate of adverse events identified (some may perceive more iatrogenesis than others).
- Third, some raters may think AEs are more common in some hospitals than others (if they can identify hospitals) leading to a hospital-rater interaction $\sigma^2_{\text{hospital x rater}}$.
- Raters may also grade some types of cases more severely than others. One reviewer may be convinced that most hospital acquired infections are iatrogenic, others that infections are usually part of an underlying disease process. This interaction between rater and patient within hospital term can not be estimated separately from the residual error term ϵ quantifying any other source of variation so they are confounded as $\sigma^2_{\text{patient:hospital x rater},\epsilon}$

The reliability calculation for using a single randomly selected medical record from each hospital, reviewed by a single randomly selected rater, to distinguish between hospitals is:

$$\text{reliability } R = \frac{\sigma^2_{\text{hospital}}}{\sigma^2_{\text{hospital}} + \underbrace{\sigma^2_{\text{patient:hospital}} + \sigma^2_{\text{rater}} + \sigma^2_{\text{hospital x rater}} + \sigma^2_{\text{patient:hospital x rater},\epsilon}}_{\text{Noise}}}$$

For our operational measurement we average over 75 patients reviewed by 2 randomly selected raters each (a total of 150 reviews per hospital) and blind the raters to hospital thus removing any possible rater x hospital interaction. Then the reliability calculation for our operational measurement is:

$$\text{reliability } R = \frac{\sigma^2_{\text{hospital}}}{\sigma^2_{\text{hospital}} + \underbrace{\frac{\sigma^2_{\text{patient:hospital}}}{75} + \frac{\sigma^2_{\text{rater}}}{2} + \frac{\sigma^2_{\text{patient:hospital x rater},\epsilon}}{2*75}}_{\text{Noise}}}$$

There are still some other strategies to improve reliability. We could do more training and attempt to reduce the rater and possibly the person x rater variance components or decide to standardize the raters. Of course the training imposes the views in the training on the sampled raters possibly limiting the generalizability of their judgement to those made by other reviewers trained differently. One might be tempted to use a reliability calculation with fixed rater effects allowing you to remove the σ^2_{rater} component. But then the reliability is not generalizable to measurements with a different set of raters. This discussion defers consideration of the fact that rater judgement about adverse events is always dichotomized in usual research practice, which complicates the reliability calculation (see Appendix 3). But the basic principles are the same.

Appendix

Appendix 1 - Reliability in the presence of a gold-standard

Barnhart provided a useful taxonomy for the different types of contexts in which researchers seek to describe characteristics of measurements.[9]

1. There is one measurement method and no gold-standard
2. There are two or more measurement methods and no gold-standard
3. There is a reference or gold-standard measurement but it is expensive, difficult or intrusive and an alternate method is proposed for comparison

In the first context, the research purpose is focused on describing reliability, or the ability of the measurement to distinguish between the objects of measurement and the precision of measurement relative to the true measured value (as defined in the text). Assessing the validity of the measurement is of necessity left to an argument based on information external to the study in which information about reliability is generated.

The second context is a complex but common one in the literature. The purpose is often described as assessing the relationship or "agreement" between methods, both of which are measured with error so as to determine if the measurements can be used interchangeably. This purpose does not directly address the quality of a measurement as it generally does not answer questions about the reliability or validity of either measurement. Usually the measurements are specified as sharing the same proposed underlying true value, although this is not always explicitly stated. This is the context for which Bland and Altman developed their popular descriptive limits of agreement methods.[39] With a design that has multiple measurements for both methods it can be thought of as two simultaneous measurement studies with the added ability to describe the relationship between the two measurements as a correlation between their predicted true measured values. When there are multiple possible alternate measurements it is more productive to categorize this context as a variation of the first context for which the measurement method is merely another source of variation in the measurement in which case the methods discussed in the main paper can be used.

In the third case, the study design is most obviously able to describe validity, or the degree to which the measurement represents the target of measurement, which in this case is the value represented by the gold standard measurement. But if there are two or more measurements are made with the alternate measurement, then the reliability as well as any systematic bias of the alternate measurement can also be assessed. This context is what I address next.

While classical test theory was the foundation of modern measurement theory the model should look a bit simplistic. When a gold-standard or reference measurement is available there are some natural extensions that make sense as we think about how a measurement made using a new-method might be different from the reference measurement. An expanded measurement model is shown below (The following discussion is adapted from one in Choudhary, 2017.[40]):

$$X = \delta + \lambda T + \epsilon$$

This looks like the classical measurement model but with the addition of two additional terms δ and λ. This expanded model has several additional components relative to the classical measurement theory model.

- δ : is the fixed bias in measurement. There is no bias if the fixed bias = 0. Imagine a scale used to weigh all the patients on a ward of a hospital having a bias of 4lb with respect to a gold standard calibrated scale. Every patient would have a weight that is 4 lbs higher than their true weight. Note that having identified and modeled this source of variation it is no longer part of the signal or the noise in the measurment. When this source of variation is not explicitly identified in the fixed part of the model and scales are sampled randomly for use in measurement than it becomes a variance component in the noise part of the measureument. The step of modeling delta as a fixed effect for a source of variation is referred to as calibration.
- λ : is the proportional scale or the rate of change in the observed measurement for a 1 unit change in the true value. There is no scale difference from the true value if the rate of change is 1 unit per 1 unit change in the true value.

- λ/σ: (where σ is the standard deviation of ϵ) is a quantity known as sensitivity which standardizes the relative rate of change of the observed measurement in terms of standard deviations of the measurement error. The larger the change in the observed measurement per unit of measurement error, the more sensitive the measurement is.
- While T is the true value of the thing you are trying to measure, $\delta + \lambda T$ can be defined as the "true" measurement value, in the sense that it is free from any random measurement error and has a single fixed value for each object of measurement (or member of a population) given a specific measurement procedure.
- This model can be expanded further to allow for heterogeneity of the error term across methods.

Using the technical definition of reliability specified above with this model, the reliability of the new measurement would be calculated as follows:

$$R = \frac{\lambda^2\tau^2}{\lambda^2\tau^2 + \sigma^2}$$

In the main text I emphasized the fact that reliability was a summary measure and here again we see in this expanded model that now three numbers representing different measurement characteristics are combined in the calculation of reliability: 1) any scale differences from the true values (λ), 2) measurement error or noise (σ^2) and 3) the distance between the objects of measurement, or the true population differences (τ^2). The signal in this case is $\lambda^2\tau^2$. The square root of the measurement error is the standard error of measurement. With some algebraic manipulation, if you divide the numerator and the denominator by σ^2 you could also have an equation for reliability that is a function of either the sensitivity as defined above (λ/σ) or a signal to noise ratio ($\lambda^2\tau^2/\sigma^2$). In order to estimate the parameters in the expanded model above generalized and linear mixed models (GLMM) are now the standard method used.[40,41]

The proportional scale parameter λ can not be estimated directly in measurement quality studies done for measurements without a gold-standard estimation as the model is not identified without the "true value" for each observation as provided by the gold-standard or reference measurement. However, measurement models in the absence of a gold-standard can be expanded to estimate a *relative* proportional scale parameter when there are multiple measurement methods or perhaps

items in a scale used per person (or object of measurement). Estimation is possible by using a model identification strategy of fixing the *lambda* parameter of one measurement method or survey item to 1 and then calculating scale parameters for the other measurements relative to the one that is fixed. When this model also allows for fixed bias and error heterogeneity it is same model as the univariate common factor model used in factor analysis.[42] Estimation of models in the absence of a gold standard that have features of the common factor model can be done with Generalized Linear and Latent Mixed Models (GLLAMMs).[32]

For binary measurements, often the kappa coefficient is used in studies comparing an alternate measurement with a gold-standard. This is using kappa not as a reliability coefficient but as a measure of association. Kraemer notes that the weighted kappa functions as a relative quantification of validity in this case and is particularly well suited for this task because of the transparency of the weighting of false negatives and false positives.[14] She notes that all 2x2 measures of association have an implicit weighting of these quantities but only the weighted kappa makes that explicit, with Fleiss' kappa being a weighted kappa with weights of 1/2 for each of false positives and negatives.

Appendix 2 - The multilevel model for reliability

The basic causal model for measurement proposes a latent true-measured value where there may be multiple observed measurements (i) of a given subject or object of measurement (j) is shown in the model in Figure 1.

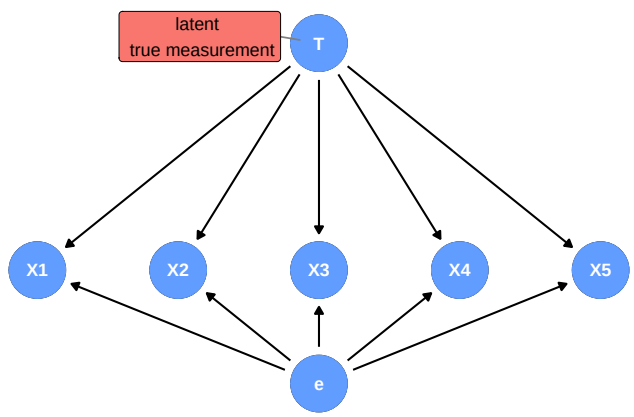

Figure 1: Multilevel measurement model

This model specifies more than just the clustering of observations within subject. It is a directed acyclic graph (dag) where we are proposing that the true value of the measurement is a latent variable (e.g. not observed) that **causes** the observed values X1-X5 (albeit with some contribution from the measurement error e). Conversely, the observed measurements are sometimes described as *reflections* of the underlying latent entity which is not directly measured but estimated from the reflected observations in the observable world. The proposal that the true measurement causes the observed measurements is inherent in classical measurement theory and all of its model extensions although it is usually not made this explicit.

Based on the causal DAG above we can propose the following multilevel statistical model.

$x_{ij} = \beta_0 + t_j + \epsilon_{ij}$

Where: β_0 is the population mean true value

x_{ij} is the ith measurement of the j-th unit

where the distribution of the true values T for the j subjects (or objects of measurement)

is represented by $t \sim N(0, \sigma_t^2)$around the true mean β_0

where $\epsilon \sim N(0, \sigma_\epsilon^2)$ represents the measurement errors that:

have the same distribution for every unit, e.g. a single variance applies to all units

are independent of the true scores

and are mutually independent $\implies$ x_{ij} are independent conditional on t_j

x_{ij} is biased if the expected mean of the measurement error is $\neq 0$

The assumptions listed for the statistical model, which are the same as those required for the classical measurement theory model shown above, define repeated parallel measurements clustered within the object of measurement, often a person in medicine. Repeated parallel measurements have the same true measurement value (β_0), no differences in scale between them, and the same measurement error distribution ($\epsilon \sim N(0, \sigma_\epsilon^2)$. This is an often used but obviously restrictive model. Fortunately it can be expanded to allow more flexible measurement models where repeated measurements are allowed to differ in their underlying true measured values, scale, and measurement error as discussed in Appendix 1.

The estimates from this model include the variance of the random effects σ_t^2 and the variance of the residuals σ_ϵ^2. With the estimates we can also predict the individual random effects representing the true measured value for each object of measurement as well as calculate a quantity known as the intraclass correlation (ICC) shown below. It should be obvious that this ICC fits the technical definition of reliability. It is more specifically the reliability of using a single measurement to distinguish between the objects of measurement .

$$\text{reliability} = R_1 = \frac{\sigma_t^2}{\sigma_t^2 + \sigma_\epsilon^2} = \text{ICC}$$

For dichotomous measurements a multilevel model can be defined as follows.

$logit(p_{ij}) = \beta_0 + t_j$

β_0 is the single true mean value over the population

x_{ij} is the ith measurement of the j-th unit

where the true value $t \sim N(0, \sigma_t^2)$

$Y \sim Binomial(n, p_{ij})$

For this model we can define the reliability on the log-odds scale in the same way as above with one slightly confusing change.

$$\text{reliability} = R_1 = \frac{\sigma_t^2}{\sigma_t^2 + \frac{\pi^2}{3}} = \text{ICC}$$

Again, this is a type of intraclass correlation coefficient that can be used to express the reliability of a dichotomous measurement. In this formula the residual variance (what is usually labeled for continuous data as σ^2) is now defined as a constant $\frac{\pi^2}{3}$. This is in fact what the residual variance is for any logistic regression. The explanation for that goes beyond the discussion of reliability here but is central to any interpretation of logistic, ordinal or multinomial regressions whether single or multilevel.[43,44]

Appendix 3 - The relationship of the latent to the manifest scale of reliability

We can illustrate the relationship of the latent to the manifest reliability while holding the variance of the true scores and the error variance constant on the latent scale in Figure 2.

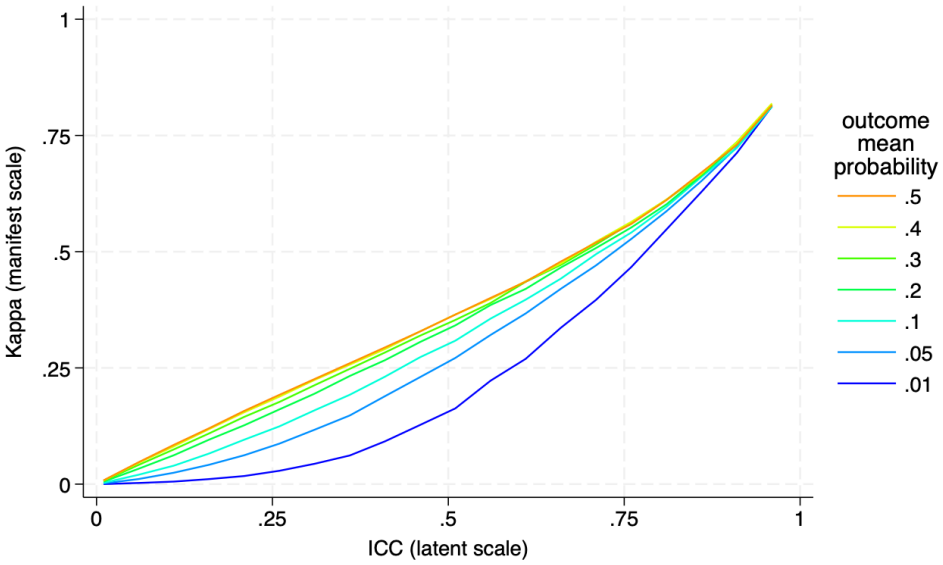

Figure 2: **Latent and manifest reliability** The y-axis shows the kappa at different levels of a latent reliability shown on the x-axis for different levels of baseline probability

These results are obtained by specifying an ICC on the latent log-odds scale along with a baseline probability and variance of the true measured latent value and then simulating multiple binomial dichotomous measurements for each member of the population from which we then calculated a kappa statistic. We can see that at all levels of baseline probability the kappa value is considerably smaller than the latent ICC for the same measurement. At lower levels of baseline probability and ICC the difference increases. These results can also be derived analytically giving the same results although the calculations of the reliabilities on the manifest scale involve complex integrals.[34,45,46]

Any analysis of binary and ordinal data has problems with interpretation that result from the two scales of the analysis, whether it involves measurement, causal inference or prediction. Reporting reliability of these type of data should where possible include estimates on both the latent and manifest scale with explicit arguments about which is the most relevant to the purpose of a specific measurement. The intraclass kappa fits the technical definition of reliability on the manifest scale and preserves many of the advantages of reliability calculated for continuous measurements. Thus it remains an important tool as other ways of estimating reliability on the manifest scale are difficult to use for applied measurement in the healthcare quality and safety fields.[34]

There are numerous off-target critiques of kappa concerning the seemingly "too low" value of kappa and the dependence of kappa on the prevalence or average probability for the population. In software coding parlance, this is a feature not a bug.[6,14,15] Kappa is dependent on prevalence because of the connection of the mean and variance in binary data (σ^2=p(1-p)). Low prevalence of events means a lower variance, a more homogeneous population and thus a lower reliability (as the numerator variance is smaller). [6] The so-called kappa paradox merely reflects the fact that measurements where the prevalence of the event are rare reflect greater homogeneity of the population.[6] Reliability should decrease in this situation.

An intuitive explanation of why kappa might have these features is that measurement on the manifest scale is a more difficult task. It requires not just locating an observation on a continuous scale, but also locating thresholds on that continuous scale with which to define the category in to which the object is placed. To give a specific example, a pathologist first places a specimen in their mind on a continuous scale of not cancer to cancer and then locates a threshold with which to categorize the specimen as cancer or not. A different pathologist has to do the same two tasks and each task offers some possibility of disagreement. The same holds for an expert clinician deciding whether an adverse event occurred when reviewing a medical record. This task is even more difficult at the extremes of the distribution when the baseline probability is small.

Appendix 4 - more on conditions of measurement

It was an effort led by Cronbach that led to the evolution of measurement theory in which the error or noise term in the denominator of the reliability calculation is thought of as being composed of multiple sources of variation in the measurement.[7] He had been funded in 1957 by NIMH to write a handbook of measurement theory and as quoted in the introduction to the resulting publication he said:

"Since reliability has been studied thoroughly and is now understood," I suggested to the team , "let us devote our first few weeks to outlining that section of the handbook, to get a feel for the undertaking." We learned humility the hard way - the enterprise never got past that topic. Not until 1972 did the book appear…that exhausted our findings on reliability"

This 15 year project, and the work of those who developed their findings farther, so expanded the implications and usefulness of the concept of reliability that they chose to call their reinterpretation "generalizability theory." Instead of seemingly different reliability coefficients with different names for things like inter and intra-rater reliability, inter-unit reliability, test-retest reliability, or sum score reliability (such as Cronbach's alpha), there is the single technical definition of reliability, the components of which differ in logical ways with the purpose and conditions of measurement. Rather than different terms like inter and intra-reliability this newer framework only requires you to specify which source of variation a measurement can be generalized over. So reliability for a measurement that can be generalized across raters represents inter-rater reliability and reliability for a measurement generalized across occasions for a single rater replaces the term intra-rater reliabilty.[27] These descriptions better specify what the reliability calculation tells you about the use of a measurement.

There are 6 key features of measurement that this expansion of classical measurement theory made more explicit and transparent:

1. The purpose or way that the measurement will be used determines the signal or numerator of the reliability calculation.
2. For the purpose of describing and improving measurement, the noise or error term is better thought of as a combination of deviations from the true measured value induced by the major sources of variation.
3. Once the reliability of using a single measurement is calculated it is both easy and transparent to calculate what the reliability would be for multiple measurements across any or all sources of variation.
4. The components included in the denominator vary depending on whether the measurement will be used as a relative or absolute measurement
5. Standardization of the conditions of measurement is reflected in the exclusion of components in the noise or error term of the denominator **and** (less intuitively) inclusion of some interaction components in the signal or numerator.
6. The design of the measurement study can restrict the identification of error components making it more difficult to make the appropriate reliability calculation.

Feature 1 to 3 are illustrated addressed in the paper and 4 and 5 are partially addressed but we go into some more detail on all of them here. I will briefly give examples of these features but for more complete descriptions researchers should consult the published primers that review these features in more extended detail.[10,27,29–31]

Prototype example

The primers of generalizability theory always start with a prototypical example where the framework can be deployed in its fullest form. I will give such an example relevant to the quality and safety literature. Imagine a hypothetical measurement system for assessing the quality of care involving standardized patients (SPs) who make unannounced visits to physicians. These visits result in measurements that have the purpose of distinguishing between physicians with respect to the quality of care that they delivered as assessed by review of a videotaped interaction for a specific presenting condition. Over the course of a year a large random selection of all visits to a physician are videotaped (to normalize the videotaping experience and obscure the SPs) but only the visits with 5 randomly selected standardized patients are stored and each taped encounter is reviewed by 5 expert raters. This results in 25 measurements per physician.

This represents a fully crossed or factorial design in which every physician is visited by every SPs and every encounter is evaluated by each rater. In the generalizability theory notation it would be a Physician x SP x Rater (or P x SP x R) design. In this case the reliability calculation has the following components:

- First, the SPs themselves may be more or less transparent in the way they manifest their presenting condition such that performance is better or worse across all physicians and raters for some SPs giving rise to a variance σ^2_{SP}
- The raters may differ from each other in the overall rate at which they judge performance (some being more severe than others in their grading) with variance σ^2_{rater}
- Third, individual raters may rate a particular physician's actions more or less favorably (something about the physicians style rubs them the wrong way) resulting in a physician-rater interaction $\sigma^2_{\text{SP x physician}}$.
- Raters may also grade the presentations of some SPs more severely than others on average. This produces a $\sigma^2_{\text{SP x rater}}$

- Finally the residual error includes any deviations resulting from factors related to a particular physican-rater-SP interaction $\sigma^2_{\text{physician x SP x rater}}$, and is confounded with any other unidentified sources of variation σ^2_ϵ and thus is represented by component identified as $\sigma^2_{\text{physician x SP x rater, }\epsilon}$.

The reliability is:

$$\text{reliability } R = \frac{\sigma^2_{\text{physician}}}{\sigma^2_{\text{physician}} + \underbrace{\frac{\sigma^2_{\text{SP}}}{5} + \frac{\sigma^2_{\text{rater}}}{5} + \frac{\sigma^2_{\text{SP x rater}}}{5\times5} + \frac{\sigma^2_{\text{physician x rater}}}{5} + \frac{\sigma^2_{\text{physician x SP}}}{5} + \frac{\sigma^2_{\text{physician x SP x rater, }\epsilon}}{5\times5}}_{\text{Noise}}}$$

This reliability calculation illustrates features 1-3 described above. Note that this design does not allow us to calculate an equivalent of an intra-rater reliability given that we can not wipe the memory of the rater, we are unable to get independent ratings by the same rater of a specific video and SP physician-encounter.

Absolute and relative measurements

The calculation above gives the reliability when using the measurement as an absolute (criterion) rather than a relative (or normative) measurement. For the former, the value for a physician must have some meaning independent of its relationship to other values. If we can define an minimum satisfactory level of performance independent of the results of the measurements that we make, then we are using an absolute measurement. If all we can do is identify the top and lowest performers relative to the others, then it is being used as a relative measurement. For the latter purpose the reliability does not depend on the σ^2_{rater} and σ^2_{SP} variance components and thus is calculated in the following way:

$$\text{reliability } R = \frac{\sigma^2_{\text{physician}}}{\sigma^2_{\text{physician}} + \underbrace{\frac{\sigma^2_{\text{SP x rater}}}{5\times5} + \frac{\sigma^2_{\text{physician x rater}}}{5} + \frac{\sigma^2_{\text{physician x SP}}}{5} + \frac{\sigma^2_{\text{physician x SP x rater, }\epsilon}}{5\times5}}_{\text{Noise}}}$$

Clearly the reliability is higher if a measurement is used only as a relative measurement as the denominator is smaller after excluding several terms (given that variances are positive). Of the examples discussed, blood pressure and adverse event rates would be most consistent with absolute measurements. The measured values can and are often interpreted independent of the relative values of a sample of values. Patient satisfaction scores have a less obvious absolute interpretation of the values. The example above might be amenable to some external determination of an absolute threshold of performance if there was an expert consensus about the minimum number of correct actions in an encounter but values could also be interpreted only as a relative measurement.

Standardized sources of variation

Standardization of measurement conditions refers to efforts to make the procedures and steps involved in measurement more consistent across testing sites and occasions and reduce the noise or the magnitude of the error variance in measurement.[47] The term "measurement standardization" developed from the use of measurement reference standards developed by international authorities

and used to define physical quantities such as units of length, mass and electrical potential.[48] In the context of generalizability theory, a "standardized measurement procedure samples observations from a sub-universe of the universe of generalization."[49] In this sense, a way to improve reliability, by making the noise component of variation in the denominator of the reliability calculation smaller, is to restrict the generalizability of the measurement to a fixed number of levels of one source of variation.

More concretely in the specific example above, we could argue that we want to generalize our measurement of physician quality to assessments made **only** by the 5 expert raters used in this study and not to any other sample of raters with similar levels of expertise. Perhaps they are the chosen representatives to a national panel of approved expert raters who will be employed going forward to assess physician quality of randomly selected physicians over the next 5 year period. In that case the reliability as an absolute measurement is as follows:

$$\text{reliability } R = \frac{\sigma^2_{\text{physician}} + \frac{\sigma^2_{\text{physician x rater}}}{5}}{\sigma^2_{\text{physician}} + \frac{\sigma^2_{\text{physician x rater}}}{5} + \underbrace{\frac{\sigma^2_{\text{SP}}}{5} + \frac{\sigma^2_{\text{SP x rater}}}{5} + \frac{\sigma^2_{\text{physician x SP}}}{5} + \frac{\sigma^2_{\text{physician x SP x rater, }\epsilon}}{5\times 5}}_{\text{Noise}}}$$

Note that we have removed the σ^2_{rater} component from the denominator and we have moved the $\sigma^2_{\text{physician x rater}}$ component from the noise to become part of the signal. This second step is somewhat less intuitive. But by stipulating that we will always use these 5 raters for all assessments of the quality of the SP-patient interaction, the person level interaction effects that result from how these specific raters evaluate evaluate them become part of the signal. Both of these changes result in an increase in the reliability of the operational measurement. However, they reduce the ability to generalize to the larger population of all possible expert raters who could evaluate physician quality.

As noted in the text, this is a measurement procedure modification that is analogous to studies that might standardize blood pressure measurement by always using a group of 5 highly calibrated blood pressure cuffs for all measurements. Another example might be a clinical trial of carotid artery surgery where the measurements done at every study site of the degree of carotid occlusion are done using carotid Doppler ultrasound measurements that are read only by 3 expert research ultrasonographers at a central coordinating center.

Measurement study design and the reliability calculation

Nested mesurement study design

The design of the measurement study can cause some of variance components to be confounded when sources of variation are not fully crossed. This can make it difficult to make the correct reliability calculation for some operational uses of the measurement. A source of variation A is crossed with B if "each level of A occurs at all levels of B".[27] If every level of B had distinct levels of B then the sources of variation would be nested with A nested within B.

Recall the example given in Table 5. In that example, patients only were admitted to one hospital and thus are considered nested within hospital rather than crossed with hospitals in a fully factorial design. There I illustrated the use of a measurement study to calculate the reliability of using

hospital average adverse event rates to distinguish between hospitals. But if we wanted to use the same measurement study to calculate the reliability of using a single measurement to distinguish between patients randomly selected from different hospitals with respect to whether they had an adverse event or not, the calculation would have the following form:

$$\text{reliability } R = \frac{\sigma^2_{\text{person:hospital}} + \sigma^2_{\text{hospital}}}{\sigma^2_{\text{person:hospital}} + \sigma^2_{\text{hospital}} + \underbrace{\sigma^2_{\text{rater}} + \sigma^2_{\text{hospital x rater}} + \sigma^2_{\text{person x rater:hospital, }\epsilon}}_{\text{Noise}}}$$

Because we have specified that we want to distinguish between patients over hospitals, and patients are nested within hospitals, the hospital variance becomes part of the differences between patients. However, the interaction term between hospital and person should then also be in the signal component when we specify that we want to distinguish between patients over hospitals.

But the nesting also results in the confounding of some of the interaction terms as follows:

$$\sigma^2_{\text{person:hospital}} = \sigma^2_{\text{person}} + \sigma^2_{\text{person x hospital}}$$

and

$$\sigma^2_{\text{person x rater:hospital}} = \sigma^2_{\text{person x rater}} + \sigma^2_{\text{person x rater x hospital}}$$

For this example, the interaction terms that are confounded within the nested variance expressions belong in the same place as the main effects. But that is not always the case. For example if we wanted to standardize or fix the raters, the $\sigma^2_{\text{person x rater}}$ component should go in the numerator as well. But we can not make that calculation as the interaction term is confounded with the residual error term. In that case the reliability that we calculate would be an underestimate. So nested designs can limit your ability to make the correct reliability reliability calculation.

Of note, there are two different reliability calculations related to distinguishing between patients. The purpose to which we put the measurement might involve distinguishing between patients **within** a hospital as opposed to distinguishing beteween patients across hospitals. In that case the reliability calculation is the following:

$$\text{reliability } R = \frac{\sigma^2_{\text{person:hospital}}}{\sigma^2_{\text{person:hospital}} + \underbrace{\sigma^2_{\text{rater}} + \sigma^2_{\text{person x rater:hospital, }\epsilon}}_{\text{Noise}}}$$

Reliability calculations for intra-rater, test-rest and inter-rater reliability

Intra-rater reliability, test-retest reliability and inter-rater reliability are names given to classical reliability calculations. In the generalizability framework these can done using a single measurement study with repeated measurements for each person, with 2 or more occasions as well as 2 or more raters or the multiple scale items within person depending on the intended reliabilty calculation.

The first two calculations, intra-rater reliability and test-retest reliability, both answer questions about reliability when generalizing across occasions at which measurements are done while rater or the scale with its component items is held constant (or is *fixed* in the terminology of gneralizability

theory). The idea is to assess reliability if a measurement is repeated while other sources of measurement error are held constant.

Intra-rater reliability

For Intra-rater reliability, we are quantifying the ability to distinguish between persons (as the usual object of measurement) when using a measurement done by the same rater at different randomly selected occasions. Thus rater is fixed and the occasion is random for this calculation.

If we are measuring something like adverse events then our goal is to make measurements on an absolute scale (e.g. what proportion of surgical cases involve an adverse event). Then the reliability is calculated as follows.

$$\text{reliability } R = \frac{\sigma^2_{\text{person}} + \sigma^2_{\text{person x rater}}}{\sigma^2_{\text{person}} + \sigma^2_{\text{person x rater}} + \underbrace{\sigma^2_{\text{occasion}} + \sigma^2_{\text{person x occasion}} + \sigma^2_{\text{person x rater x occasion, }\epsilon}}_{\text{Noise}}}$$

As rater is fixed the main effect variance drops out of the denominator and the interaction becomes part of the signal and is added to the numerator. If the measurement was to be used only as a relative measurement with everyone measured at the same or effectively the same occasions then the main effect varance for occasion would also be dropped out of the denominator. To make this calculation you need variance components that are produced from a measurement study with raters, person and occasions in a fully crossed design. If any interaction is not modeled explicitly, the interaction variance collapses into the residual term [50]. In that case you would not be able to make the reliability calculation correctly as the $\sigma^2_{\text{person x rater}}$ would be hidden in the residual variance.

Test-retest reliability

Test-retest reliability is the same calculation but would be used to quantify the ability of a measurement such as patient satisfaction to distinguish between persons when using a measurement is done with the same set of items or scale at two different occasions. As the measurement averages over n_i items to form the scale the variance components involving items are divided by the number of items in the scale. You can just thus substitute $\frac{1}{n_i}\sigma^2_{\text{item}}$ for σ^2_{rater} in the above equation for intra-rater reliability to get:

$$\text{reliability } R = \frac{\sigma^2_{\text{person}} + \frac{1}{n_i}\sigma^2_{\text{person x item}}}{\sigma^2_{\text{person}} + \frac{1}{n_i}\sigma^2_{\text{person x item}} + \underbrace{\sigma^2_{\text{occasion}} + \frac{1}{n_i}\sigma^2_{\text{person x occasion}} + \frac{1}{n_i}\sigma^2_{\text{person x item x occasion, }\epsilon}}_{\text{Noise}}}$$

In this case the n_i scale items are fixed so the variance of the interaction of person with scale items adds to the signal variance and the variance of the main effects of item are removed from the denominator. Again if the measurement was to be used only as a relative measurement with everyone measured at the same or effectively the same occasions then the main effect varance for occasion would also be dropped out of the denominator.

Inter-rater reliability

Inter-rater reliability is used to quantify the ability of a measurement to distinguish between persons when using a measurement done using a randomly selected reviewer on fixed occasions. Many classical reliability coefficients for the inter-rater reliability calculate reliablilty at a single occasion. This requires fixing the occasion and again moving the variance component of the interaction into the numerator while the variance of the main effect of occasion is removed. In this way the reliability is calculating the ability to distinguish between people with a random rater while holding other sources of variation (occasion) constant. Brennan refers to this as a standardized interrater reliability and the formula is shown below.[51]

$$\text{reliability } R = \frac{\sigma^2_{\text{person}} + \sigma^2_{\text{person x occasion}}}{\sigma^2_{\text{person}} + \sigma^2_{\text{person x occasion}} + \underbrace{\sigma^2_{\text{rater}} + \sigma^2_{\text{person x rater}} + \sigma^2_{\text{person x rater x occasion, }\epsilon}}_{\text{Noise}}}$$

If interaction terms are not calculated when using a multilevel model to calculate reliability, then the $\sigma^2_{\text{person x occasion}}$ term would collapse into the variance of the main effect of occasion. The calculated reliability would then only be an approximation of the intended reliability calculation for the inter-rater reliability at a single occasion.

A potentially more relevant reliability

An alternative and possibly more appropriate calculation than the more specific reliability calculations demonstrated above would look at reliability when using a measurement based on a randomly selected rater at a randomly selected occasion. This seems like a more typical common measurement procedure (rarely do all raters do all their work at the same time) and it is relevant for generalizing the measurement to other people, raters and occasions that can be considered as utilizing additional samples from the same populations of those used for the measurement study. The reliability calculation would then be simpler:

$$\text{reliability } R = \frac{\sigma^2_{\text{person}}}{\sigma^2_{\text{person}} + \underbrace{\sigma^2_{\text{occasion}} + \sigma^2_{\text{person x occasion}} + \sigma^2_{\text{rater}} + \sigma^2_{\text{person x rater}} + \sigma^2_{\text{person x rater x occasion, }\epsilon}}_{\text{Noise}}}$$

If the interaction terms are not calculated the associated variances will collapse into the variance of the main effect for rater and occasion. This will not be a problem for the reliability calculation. Note that for both of these calculations you need a fully factorial measurement study to estimate these variance components which would require raters to rate the same person on two occasions which is usually not feasible. If that is not possible due to contamination of the raters judgement by the earlier rating session, then you end up with a nested design with its associated limitations as described above. See Brennan 2003[51] for more details and worked examples.

Partially crossed measurement study designs

Other measurement study designs can limit the reliability calculations that can be made. In the context of the standardized patient measurement study described above, it is common that we can not arrange for every physician to be evaluated by every rater and SP. In fact a more common design is just to have a random selection of raters and SPs for each physician. This forms what is

called a "partially crossed" design. The farther away from a fully crossed design that you get, the more difficult it gets to estimate the interaction terms and if you can not estimate an interaction it becomes part of the residual variance.[50] Thus a sparsely crossed design may also result in interaction variance components that are confounded and impossible to move for the purpose of the reliability calculation independent of the terms with which they are confounded.

Appendix References

1 Shaw PA, Gustafson P, Carroll RJ, *et al.* STRATOS guidance document on measurement error and misclassification of variables in observational epidemiology: Part 2—More complex methods of adjustment and advanced topics. *Statistics in Medicine*. 2020;39:2232–63. doi: 10.1002/sim.8531

2 Keogh RH, Shaw PA, Gustafson P, *et al.* STRATOS guidance document on measurement error and misclassification of variables in observational epidemiology: Part 1—Basic theory and simple methods of adjustment. *Statistics in Medicine*. 2020;39:2197–231. doi: 10.1002/sim.8532

3 Jacoby WG. Levels of Measurement and Political Research: An Optimistic View. *American Journal of Political Science*. 1999;43:271–301. doi: 10.2307/2991794

4 Hand DJ. Statistics and the Theory of Measurement. *Journal of the Royal Statistical Society Series A (Statistics in Society)*. 1996;159:445–92. doi: 10.2307/2983326

5 Campbell NR. *Physics - the elements*. Cambridge: Univ. Press 1920.

6 Streiner DL, Norman GR, Cairney J. *Health measurement scales: A practical guide to their development and use*. Fifth edition. Oxford: Oxford University Press 2015.

7 Cronbach LJ, editor. *The Dependability of behavioral measurements: Theory of generalizability for scores and profiles*. New York: Wiley 1972.

8 Lord FM, Novick MR, Lord FM, *et al.* *Statistical theories of mental test scores*. Reading, Mass.: Addison-Wesley Pub. Co. 1968.

9 Barnhart HX. A Review on Assessing Agreement. *Wiley StatsRef: Statistics Reference Online*. John Wiley & Sons, Ltd 2018:1–30.

10 Brennan RL. *Generalizability theory*. New York: Springer 2001.

11 Strauss ME, Smith GT. Construct validity: Advances in theory and methodology. *Annual Review of Clinical Psychology*. 2009;5:1–25. doi: 10.1146/annurev.clinpsy.032408.153639

12 Norman GR. Discrimination and reliability: Equal partners? *Health and Quality of Life Outcomes*. 2008;6:81. doi: 10.1186/1477-7525-6-81

13 Kraemer HC. Measurement of reliability for categorical data in medical research. *Statistical Methods in Medical Research*. 1992;1:183–99.

14 Kraemer HC, Periyakoil VS, Noda A. Kappa coefficients in medical research. *Statistics in Medicine*. 2002;21:2109–29. doi: 10.1002/sim.1180

15 Vach W. The dependence of Cohen's kappa on the prevalence does not matter. *Journal of Clinical Epidemiology*. 2005;58:655–61. doi: 10.1016/j.jclinepi.2004.02.021

16 Vach W, Gerke O. Gwet's AC1 is not a substitute for Cohen's kappa - A comparison of basic properties. *MethodsX*. 2023;10:102212. doi: 10.1016/j.mex.2023.102212

17 Vangeneugden T, Laenen A, Geys H, *et al.* Applying Concepts of Generalizability Theory on Clinical Trial Data to Investigate Sources of Variation and Their Impact on Reliability. *Biometrics*. 2005;61:295–304. doi: 10.1111/j.0006-341X.2005.031040.x

18 Laenen A, Vangeneugden T, Geys H, *et al.* Generalized reliability estimation using repeated measurements. *The British Journal of Mathematical and Statistical Psychology*. 2006;59:113–31. doi: 10.1348/000711005X66068

19 Jiang Z, Skorupski W. A Bayesian approach to estimating variance components within a multivariate generalizability theory framework. *Behavior Research Methods*. 2018;50:2193–214. doi: 10.3758/s13428-017-0986-3

20 Allen MJ, Yen WM. *Introduction to measurement theory*. Reissued. Long Grove, Ill: Waveland Press 2002.

21 Kraemer HC. The Reliability of Clinical Diagnoses: State of the Art. *Annual Review of Clinical Psychology*. 2014;10:111–30. doi: 10.1146/annurev-clinpsy-032813-153739

22 Spearman C. Correlation calculated from faulty data. *British Journal of Psychology, 1904-1920*. 1910;3:271–95.

23 Brown W. Some experimental results in the correlation of mental abilities. *British Journal of Psychology*. 1910;3:296–322.

24 Llabre MM, Ironson GH, Spitzer SB, *et al.* How many blood pressure measurements are enough? An application of generalizability theory to the study of blood pressure reliability. *Psychophysiology*. 1988;25:97–106.

25 Gagnier JJ, Lai J, Mokkink LB, *et al.* COSMIN reporting guideline for studies on measurement properties of patient-reported outcome measures. *Quality of Life Research: An International Journal of Quality of Life Aspects of Treatment, Care and Rehabilitation*. 2021;30:2197–218. doi: 10.1007/s11136-021-02822-4

26 Kottner J, Audige L, Brorson S, *et al.* Guidelines for Reporting Reliability and Agreement Studies (GRRAS) were proposed. *International Journal of Nursing Studies.* 2011;48:661–71. doi: 10.1016/j.ijnurstu.2011.01.016

27 Bloch R, Norman G. Generalizability theory for the perplexed: A practical introduction and guide: AMEE Guide No. 68. *Medical Teacher.* 2012;34:960–92. doi: 10.3109/0142159X.2012.703791

28 Satpathy S, Wundaville LT, Satapathy S, *et al.* A Systematic Review of Patient Satisfaction Scales and Their Applicability to Covid-19 Hospitalized Patients: Gaps and Emerging Needs. *Journal of Patient Experience.* 2022;9:1–10. doi: 10.1177/23743735221079132

29 Shavelson RJ, Webb NM, Shavelson RJ, *et al.* *Generalizability theory : A primer.* Newbury Park, Calif.: Sage Publications 1991.

30 Webb NM, Shavelson RJ, Haertel EH. 4 Reliability Coefficients and Generalizability Theory. In: Sinharay CRR and S, ed. *Handbook of Statistics.* Elsevier 2006:81–124.

31 Mokkink LB, Eekhout I, Boers M, *et al.* Studies on Reliability and Measurement Error of Measurements in Medicine – From Design to Statistics Explained for Medical Researchers. *Patient Related Outcome Measures.* 2023;14:193–212. doi: 10.2147/PROM.S398886

32 Rabe-Hesketh S, Skrondal A. Classical latent variable models for medical research. *Statistical Methods in Medical Research.* 2008;17:5–32. doi: 10.1177/0962280207081236

33 Kane M. The Errors of Our Ways. *Journal of Educational Measurement.* 2011;48:12–30.

34 Milanzi E, Molenberghs G, Alonso A, *et al.* Reliability measures in item response theory: Manifest versus latent correlation functions. *The British Journal of Mathematical and Statistical Psychology.* 2015;68:43–64. doi: 10.1111/bmsp.12033

35 Kraemer HC. Kappa Coefficient. In: Balakrishnan N, Colton T, Everitt B, *et al.*, eds. *Wiley StatsRef: Statistics Reference Online.* Wiley 2015:1–4.

36 Abel GA, Agniel D, Elliott MN. The good, the bad and the ugly: What do we really do when we identify the best and the worst organisations? *BMJ Quality & Safety.* 2025;34:53–61. doi: 10.1136/bmjqs-2023-017039

37 Hiatt HH, Barnes BA, Brennan TA, *et al.* A study of medical injury and medical malpractice. *N Engl J Med.* 1989;321:480–4.

38 Bravo G, Potvin L. Estimating the reliability of continuous measures with Cronbach's alpha or the intraclass correlation coefficient: Toward the integration of two traditions. *Journal of Clinical Epidemiology.* 1991;44:381–90.

39 Altman DG, Bland JM. Measurement in Medicine: The Analysis of Method Comparison Studies. *Journal of the Royal Statistical Society Series D (The Statistician)*. 1983;32:307–17. doi: 10.2307/2987937

40 Choudhary PK, Nagaraja HN. *Measuring agreement: Models, methods, and applications*. Hoboken, NJ: John Wiley & Sons 2017.

41 Wang W, Lin N, Oberhaus JD, *et al.* Assessing method agreement for paired repeated binary measurements administered by multiple raters. *Statistics in Medicine*. 2020;39:279–93. doi: 10.1002/sim.8398

42 Skrondal A, Rabe-Hesketh S. *Generalized Latent Variable Modeling: Multilevel, Longitudinal, and Structural Equation Models*. Boca Raton: Chapman & Hall/CRC 2004.

43 Mood C. Logistic Regression: Why We Cannot Do What We Think We Can Do, and What We Can Do About It. *European Sociological Review*. 2009;26:67–82. doi: 10.1093/esr/jcp006

44 Buis ML. Logistic regression: Why we often can do what we think we can do. 2015.

45 Kraemer HC. Ramifications of a population model for κ as a coefficient of reliability. *Psychometrika*. 1979;44:461–72. doi: 10.1007/BF02296208

46 Rodriguez G, Elo I. Intra–class correlation in random–effects models for binary data - The Stata Journal Volume 3 Number 1, St0031. *Stata Journal*. 2003;3:34–46.

47 Psychological testing. *Wikipedia*. 2025.

48 Standard (metrology). *Wikipedia*. 2025.

49 Kane MT. A Sampling Model for Validity. *Applied Psychological Measurement*. 1982;6:125–60. doi: 10.1177/014662168200600201

50 Schielzeth H, Nakagawa S. Nested by design: Model fitting and interpretation in a mixed model era. *Methods in Ecology and Evolution*. 2013;4:14–24. doi: 10.1111/j.2041-210x.2012.00251.x

51 Brennan RL. Coefficients and Indices in Generalizability Theory. University of Iowa, Iowa City, IA: Center for Advanced Studies in Measurement and Assessment 2003, August.